

Solid & Hazardous Waste

FE Environmental — Day 12

Study set • 2026-06-28 • 20 questions • every numeric answer recomputed in Python

Handbook fidelity. All formulas confirmed against NCEES FE Reference Handbook v10.6 text extract (materials/NCEES.FE.Handbook.text.txt). Clay liner leachate breakthrough time $t = d^2\eta/[K(d+h)]$ on p.333 — confirmed (USCS: d in ft, K in ft/yr, h in ft). Overburden pressure formula $SW_p = SW_i + p/(a+bp)$ on p.333 — confirmed; a in yd^3/in^2 , b in yd^3/lb , p in lb/in^2 , SW_p in lb/yd^3 . Landfill gas flux $NA = D \cdot \eta_{\text{gas}} \cdot (C_{\text{fill}} - C_{\text{atm}})/L$ on p.333 — OCR garbles notation; confirmed as Fick's law modified form with gas-filled porosity η_{gas} ; D values for methane ($0.20 \text{ cm}^2/\text{s}$) and CO_2 ($0.13 \text{ cm}^2/\text{s}$) tabulated same page. Landfill cover water balance $\Delta S_{LC} = P - R - ET - \text{PER}_{\text{sw}}$ on p.333 — confirmed (all in inches). Volume reduction % = $(V_i - V_f)/V_i \times 100$ on p.334 — confirmed. Effective half-life $1/\tau_e = 1/\tau_r + 1/\tau_b$ on p.335 — confirmed. Radioactive decay $N = N_0 \exp(-0.693 t/\tau)$ on p.335 — confirmed; also written as $N = N_0 e^{(-0.693t/\tau)}$. Inverse square law $I_1/I_2 = (R_2/R_1)^2$ on p.335 — confirmed (intensity inversely proportional to square of distance). Coefficient of variation $CV = (100 \cdot s)/\bar{x}$ on p.336 — confirmed. DQO sampling table (one-sided one-sample t-test) on p.336 — confirmed; CV=15%, power=95%, confidence=95%, MDRD=20% $\rightarrow n=8$. Chemical Compatibility Chart on p.26 — confirmed; reactivity codes: H=heat generation, F=fire, GT=toxic gas generation, GF=flammable gas, E=explosion, P=polymerization, S=solubilization, G=innocuous non-flammable gas. Group 1 (Acids, Minerals, Non-Oxidizing) + Group 10 (Caustics) $\rightarrow$ H. Group 3 (Acids, Organic) + Group 7 (Amines, Aliphatic & Aromatic) $\rightarrow$ H + GT. MSW moisture content table on p.332 — confirmed (food wastes 70%, paper 6%, glass 2%, wood 20% typical). MSW heating value table on p.359 — confirmed (food 2,000 Btu/lb, paper 7,200, plastics 14,000 typical). MSW as-received density table on p.332 — confirmed (aluminum cans whole 50 lb/yd^3 , baled 950 lb/yd^3). NOT in Handbook (supplemental, tagged accordingly): RCRA four characteristics (ignitability/corrosivity/reactivity/toxicity) are regulatory knowledge from RCRA — tagged fundamental. RCRA 99.99% DRE standard for POHCs is regulatory (40 CFR 63.1206) — tagged fundamental. Diversion rate = recycled/total generated is a definitional ratio — tagged fundamental. Landfill airspace consumption = $(\text{tons}/\text{day} \times 2000 \text{ lb}/\text{ton}) / \text{density}$ is fundamental mass balance — tagged fundamental. TCLP threshold (5.0 mg/L Pb) stated in Q18 stem — tagged in-stem.

Q1. [D12-Q01 • MCQ • Landfill — clay liner leachate breakthrough time (USCS)]

A municipal landfill is lined with a 2-ft clay layer (porosity $\eta = 0.25$, hydraulic conductivity $K = 0.10$ ft/yr). The hydraulic head above the liner is $h = 3$ ft. The time for leachate to penetrate through the clay liner is most nearly:

Break-Through Time (Handbook p.333): $t = d^2\eta / [K(d + h)]$ where d = liner thickness (ft), η = porosity, K = hydraulic conductivity (ft/yr), h = hydraulic head (ft).

Relevant: $t = \frac{d^2\eta}{K(d + h)}$

- (A) 1.0 yr
- (B) **2.0 yr** ✓
- (C) 5.0 yr
- (D) 12.5 yr

Answer: (B)

$$t = \frac{(2 \text{ ft})^2 \times 0.25}{0.10 \text{ ft/yr} \times (2 + 3) \text{ ft}} = \frac{4 \times 0.25}{0.10 \times 5} = \frac{1.00}{0.50} = 2.0 \text{ yr}$$

Option (A) 1.0 yr: uses d (not d^2) in numerator $\rightarrow t = 2 \times 0.25 / (0.10 \times 5) = 1.0$ yr — square the liner thickness.
 Option (C) 5.0 yr: omits the head h from the denominator $\rightarrow t = 4 \times 0.25 / (0.10 \times 2) = 5.0$ yr — the hydraulic head drives flow through the liner and must be included.
 Option (D) 12.5 yr: uses $(d+h)^2$ in numerator instead of $d^2 \rightarrow 25 \times 0.25 / (0.10 \times 5) = 12.5$ yr. The formula penalizes thin liners quadratically and rewards greater head height — both increase leachate flux through the liner.

Handbook: Environmental Engineering — Landfill: Break-Through Time for Leachate to Penetrate a Clay Liner, $t = d^2\eta/[K(d+h)]$ (p.333) • Handbook p.333 — Landfill: Leachate Breakthrough Time

Q2. [D12-Q02 • MCQ • MSW composition — weighted average moisture content (SI)]

A solid waste stream consists of three components: 50% food wastes, 30% paper, and 20% glass (all by wet mass). Using typical moisture content values from the Handbook (food wastes 70%, paper 6%, glass 2%), the weighted average moisture content of this mixed solid waste is most nearly:

Weighted average = Σ (mass fraction $\times$ moisture content)

Relevant: $\overline{MC} = \sum_i f_i \cdot MC_i$

- (A) 26.0%
- (B) 35.0%
- (C) **37.2%** ✓
- (D) 70.0%

Answer: (C)

$$\overline{MC} = 0.50 \times 70\% + 0.30 \times 6\% + 0.20 \times 2\% = 35.0 + 1.8 + 0.4 = 37.2\%$$

Option (A) 26.0%: unweighted average $(70+6+2)/3 = 26.0\%$ — ignores the actual mass fractions. Option (B) 35.0%: counts only the food waste contribution (0.50×70) and omits paper and glass — partial calculation. Option (D) 70.0%: reports only the food waste typical moisture content without weighting — the dominant component's value, not the blend. The weighted average properly weights each component by its mass fraction; food wastes dominate because of their high fraction AND high moisture content.

Handbook: Environmental Engineering — Landfill: Typical Moisture Content of MSW Components table (p.332) • Handbook p.332 — Landfill: MSW Moisture Content Table (food wastes 70%, paper 6%, glass 2% typical)

Q3. [D12-Q03 • MCQ • Landfill — methane gas flux through cover (SI)]

Methane migrates upward through a landfill soil cover. The cover depth is $L = 60$ cm, and the gas-filled porosity is $\eta_{\text{gas}} = 0.30$. The methane diffusion coefficient is $D = 0.20$ cm²/s (from Handbook p.333). The methane concentration at the bottom of the cover (landfill side) is $C_{\text{fill}} = 2.5 \times 10^{-4}$ g/cm³, and the atmospheric concentration is $C_{\text{atm}} \approx 0$. The steady-state methane gas flux is most nearly:

Gas Flux (Handbook p.333): $N_A = D \cdot \eta_{\text{gas}} \cdot (C_{\text{fill}} - C_{\text{atm}}) / L$

Relevant: $N_A = \frac{D \cdot \eta_{\text{gas}} \cdot (C_{\text{fill}} - C_{\text{atm}})}{L}$

- (A) 8.3×10^{-7} g/(cm²·s)
- (B) **2.5×10^{-7} g/(cm²·s) ✓**
- (C) 2.5×10^{-5} g/(cm²·s)
- (D) 9.0×10^{-4} g/(cm²·s)

Answer: (B)

$$N_A = \frac{0.20 \times 0.30 \times (2.5 \times 10^{-4} - 0)}{60} = \frac{0.060 \times 2.5 \times 10^{-4}}{60} = \frac{1.5 \times 10^{-5}}{60} = 2.5 \times 10^{-7} \text{ g}/(\text{cm}^2 \cdot \text{s})$$

Option (A) 8.3×10^{-7} : omits η_{gas} — uses $N_A = D \cdot \Delta C / L = 0.20 \times 2.5 \times 10^{-4} / 60 = 8.33 \times 10^{-7}$; the gas-filled porosity reduces the effective diffusivity. Option (C) 2.5×10^{-5} : uses $L = 0.60$ cm (treats as meters); correct units require L in cm (60 cm, not 0.60 m). Option (D) 9.0×10^{-4} : multiplies L into numerator instead of dividing $\rightarrow 0.20 \times 0.30 \times 2.5 \times 10^{-4} \times 60$. Fick's first law modified by gas-filled porosity governs landfill gas transport through the unsaturated cover.

Handbook: Environmental Engineering — Landfill: Gas Flux $N_A = D \cdot \eta_{\text{gas}} \cdot (C_{\text{Aatm}} - C_{\text{Afill}}) / L$; $D(\text{methane}) = 0.20$ cm²/s, $D(\text{CO}_2) = 0.13$ cm²/s (p.333) • Handbook p.333 — Landfill: Gas Flux (Fick's Law with gas-filled porosity)

Q4. [D12-Q04 • Numeric • Landfill cover — water balance, solve for percolation (USCS)]

A landfill cover water balance for a one-year period shows: precipitation $P = 42$ in, surface runoff $R = 16$ in, evapotranspiration $ET = 18$ in, and change in soil moisture storage $\Delta S_{LC} = +2$ in. Determine the annual percolation through the landfill cover into the waste (PER_{sw} , in inches per year).

Soil Landfill Cover Water Balance (Handbook p.333): $\Delta S_{LC} = P - R - ET - PER_{sw}$

$$\text{Relevant: } \Delta S_{LC} = P - R - ET - PER_{sw} \quad PER_{sw} = P - R - ET - \Delta S_{LC}$$

Numeric entry.

Answer: 6 in/yr

Rearrange the water balance for the unknown:

$$PER_{sw} = P - R - ET - \Delta S_{LC} = 42 - 16 - 18 - 2 = 6 \text{ in/yr}$$

Every unit of precipitation must go somewhere: runoff off the surface, evapotranspiration back to the atmosphere, storage change in the soil, or percolation downward into the waste. With 6 in/yr percolating through the cover, landfill leachate generation will be proportional to this flux and the waste footprint area. Minimizing PER_{sw} through good cover design (adequate ET, runoff management) is central to leachate control.

Handbook: Environmental Engineering — Landfill: Soil Landfill Cover Water Balance $\Delta S_{LC} = P - R - ET - PER_{sw}$ (p.333) • Handbook p.333 — Landfill: Soil Cover Water Balance

Q5. [D12-Q05 • MCQ • Solid waste — volume reduction by compaction (USCS)]

A packer truck compresses uncompacted residential solid waste from an initial volume of $V_i = 800$ yd^3 to a final volume of $V_f = 200$ yd^3 . The volume reduction achieved by compaction is most nearly:

Volume Reduction (Handbook p.334): $VR(\%) = (V_i - V_f) / V_i \times 100$

$$\text{Relevant: } VR(\%) = \frac{V_i - V_f}{V_i} \times 100$$

- (A) 25%
- (B) 33.3%
- (C) **75% ✓**
- (D) 300%

Answer: (C)

$$VR = \frac{800 - 200}{800} \times 100 = \frac{600}{800} \times 100 = 75\%$$

Option (A) 25%: computes $V_f/V_i = 200/800 = 25\%$ — this is the fraction remaining (compaction ratio⁻¹), not the reduction. Option (B) 33.3%: computes $V_f/(V_i - V_f) = 200/600 = 33.3\%$ — wrong denominator. Option (D) 300%: uses $(V_i - V_f)/V_f \times 100 = 600/200 = 300\%$ — inverted; by definition, volume reduction is normalized to the original volume. A 75% VR means the packer truck reduces 4 yd^3 of loose waste to 1 yd^3 of compacted waste (4:1 compaction ratio).

Handbook: Environmental Engineering — Compaction: Volume Reduction(%) = $(V_i - V_f)/V_i \times 100$ (p.334) • Handbook p.334 — Compaction: Volume Reduction formula

Q6. [D12-Q06 • MCQ • Landfill — effect of overburden pressure on waste specific weight (USCS)]

The initial compacted specific weight of municipal solid waste in a landfill is $SW_i = 1,000 \text{ lb/yd}^3$. At an overburden pressure of $p = 10 \text{ lb/in}^2$, the empirical constants are $a = 0.008 \text{ yd}^3/\text{in}^2$ and $b = 0.00053 \text{ yd}^3/\text{lb}$. The specific weight of the waste under this overburden is most nearly:

Effect of Overburden Pressure (Handbook p.333): $SW_p = SW_i + p / (a + b \cdot p)$ where SW_p (lb/yd^3), SW_i (lb/yd^3), p (lb/in^2).

Relevant: $SW_p = SW_i + \frac{p}{a + b \cdot p}$

- (A) 752 lb/yd^3
- (B) 1,010 lb/yd^3
- (C) **1,750 lb/yd^3 ✓**
- (D) 2,000 lb/yd^3

Answer: (C)

$$SW_p = 1000 + \frac{10}{0.008 + 0.00053 \times 10} = 1000 + \frac{10}{0.008 + 0.0053} = 1000 + \frac{10}{0.0133} = 1000 + 752 = 1,752 \text{ lb/yd}^3$$

Option (A) 752 lb/yd^3 : forgets to add SW_i — reports only the overburden increment $p/(a+bp)$. Option (B) 1,010 lb/yd^3 : adds p directly to SW_i ($SW_i + p = 1000+10$) — ignores the empirical denominator. Option (D) 2,000 lb/yd^3 : rounds to the upper end of the typical range (1,750–2,150 lb/yd^3) without calculation. The typical initial compacted $SW_i \approx 1,000 \text{ lb/yd}^3$ increases to 1,750–2,150 lb/yd^3 under overburden — settlements in old landfills are driven by this densification.

Handbook: Environmental Engineering — Landfill: Effect of Overburden Pressure $SW_p = SW_i + p/(a+bp)$ (p.333) • Handbook p.333 — Landfill: Effect of Overburden Pressure on Specific Weight

Q7. [D12-Q07 • MCQ • Radioactive waste — effective half-life (SI)]

Iodine-131 (^{131}I) has a radioactive half-life $\tau_r = 8$ days. In the human body, its biological half-life is $\tau_b = 120$ days (rate of biological excretion). The effective half-life of ^{131}I in the body is most nearly:

Effective Half-Life (Handbook p.335): $1/\tau_e = 1/\tau_r + 1/\tau_b$

Relevant: $\frac{1}{\tau_e} = \frac{1}{\tau_r} + \frac{1}{\tau_b}$

- (A) **7.5 days** ✓
- (B) 8.0 days
- (C) 64 days
- (D) 128 days

Answer: (A)

$$\frac{1}{\tau_e} = \frac{1}{8} + \frac{1}{120} = \frac{15}{120} + \frac{1}{120} = \frac{16}{120} \implies \tau_e = \frac{120}{16} = 7.5 \text{ days}$$

Equivalently: $\tau_e = \tau_r \cdot \tau_b / (\tau_r + \tau_b) = 8 \times 120 / (8 + 120) = 960 / 128 = 7.5$ days.

Option (B) 8.0 days: uses only the radioactive half-life, ignoring biological clearance. Option (C) 64 days: simple arithmetic average $(\tau_r + \tau_b)/2 = 64$ — averaging is incorrect; the two parallel decay channels add as reciprocals. Option (D) 128 days: simple sum $\tau_r + \tau_b$ — if they added, the effective half-life would be LONGER than either; instead, two simultaneous clearance pathways make the effective half-life SHORTER. τ_e is always shorter than the smaller of τ_r and τ_b .

Handbook: Environmental Engineering — Radiation: Effective Half-Life $1/\tau_e = 1/\tau_r + 1/\tau_b$ (p.335) • Handbook p.335
— Radiation: Effective Half-Life

Q8. [D12-Q08 • MCQ • Radioactive waste — radioactive decay after multiple half-lives (SI)]

A hospital receives 1,000 μCi of Iodine-131 (radioactive half-life $\tau = 8$ days). After 24 days of decay in a shielded storage cabinet, the remaining activity is most nearly:

Half-Life Decay (Handbook p.335): $N = N_0 \cdot e^{(-0.693 t/\tau)}$

Relevant: $N = N_0 \cdot e^{-0.693 t/\tau}$

- (A) 62.5 μCi
- (B) **125 μCi ✓**
- (C) 250 μCi
- (D) 500 μCi

Answer: (B)

24 days = 3 half-lives ($24/8 = 3$):

$$N = 1000 \times e^{-0.693 \times 3} = 1000 \times e^{-2.079} = 1000 \times \frac{1}{8} = 125 \mu\text{Ci}$$

After each half-life: 1,000 $\rightarrow$ 500 $\rightarrow$ 250 $\rightarrow$ 125 μCi .

Option (A) 62.5 μCi : uses 4 half-lives ($t=32$ d, not 24 d) — $N = 1000/16 = 62.5 \mu\text{Ci}$. Option (C) 250 μCi : uses 2 half-lives ($t=16$ d) — $N = 1000/4 = 250 \mu\text{Ci}$. Option (D) 500 μCi : uses 1 half-life ($t=8$ d) — $N = 1000/2 = 500 \mu\text{Ci}$. The $e^{(-0.693t/\tau)}$ form and the successive-halving form are equivalent: after n half-lives $N = N_0/2^n$. Radioactive waste held for 10 half-lives drops to $N_0/1024$; 0.1% of original — this guides storage-then-disposal protocols for short-lived medical isotopes.

Handbook: Environmental Engineering — Radiation: Half-Life Decay $N = N_0 e^{(-0.693 t/\tau)}$ (p.335) • Handbook p.335 — Radiation: Half-Life Decay Equation

Q9. [D12-Q09 • MCQ • Radiation — inverse square law for intensity (USCS)]

A radiation survey meter reads 20 mrem/hr at a distance of $R_1 = 1$ ft from a point source. The radiation intensity at a distance of $R_2 = 4$ ft from the same source is most nearly:

Inverse Square Law (Handbook p.335): $I_1/I_2 = (R_2/R_1)^2$ or equivalently: $I_2 = I_1 \times (R_1/R_2)^2$

Relevant: $\frac{I_1}{I_2} = \left(\frac{R_2}{R_1}\right)^2$

- (A) **1.25 mrem/hr** ✓
- (B) 5.0 mrem/hr
- (C) 20 mrem/hr
- (D) 80 mrem/hr

Answer: (A)

$$I_2 = I_1 \times \left(\frac{R_1}{R_2}\right)^2 = 20 \times \left(\frac{1}{4}\right)^2 = 20 \times \frac{1}{16} = 1.25 \text{ mrem/hr}$$

Moving $4\times$ farther reduces intensity by $4^2 = 16\times$.

Option (B) 5.0 mrem/hr: uses a linear (not squared) relationship — $I_2 = I_1 \times (R_1/R_2) = 20 \times (1/4) = 5$. Option (C) 20 mrem/hr: assumes intensity is constant with distance (no inverse-square attenuation). Option (D) 80 mrem/hr: applies direct proportion in the wrong direction — $I_2 = I_1 \times (R_2/R_1) = 20 \times 4 = 80$. The inverse square law applies to point sources radiating isotropically: doubling the distance quarters the dose rate. This is the basis for the distance rule in radiation protection ('time, distance, shielding').

Handbook: Environmental Engineering — Radiation: Inverse Square Law $I_1/I_2 = (R_2/R_1)^2$, also written Flux at distance 2 = Flux at distance 1 $\times (r_1/r_2)^2$ (p.335) • Handbook p.335 — Radiation: Inverse Square Law

Q10. [D12-Q10 • MCQ • Hazardous waste — chemical compatibility (acid + caustic)]

A hazardous waste contractor accidentally mixes a drum of non-oxidizing mineral acid (Group 1 — e.g., hydrochloric acid) with a drum of a caustic (Group 10 — e.g., sodium hydroxide). According to the NCEES Chemical Compatibility Chart (Handbook p.26), what is the primary hazard from this mixture?

Reactivity codes: H = Heat generation · F = Fire · G = Innocuous & non-flammable gas · GT = Toxic gas · GF = Flammable gas · E = Explosion · P = Polymerization

- (A) E — Explosion
- (B) F — Fire
- (C) **H — Heat generation ✓**
- (D) GT — Toxic gas generation

Answer: (C)

From the Chemical Compatibility Chart (p.26): Group 1 (Acids, Minerals, Non-Oxidizing) × Group 10 (Caustics) → **H (Heat generation)**.

Acid–base neutralization: $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O} + \text{heat}$. This is highly exothermic — mixing a strong acid with a concentrated caustic can cause boiling, spattering, or container failure from steam pressure. Option (A) Explosion: not the primary hazard for simple acid–caustic mixing (explosion codes E appear for reactive/peroxide-forming combinations). Option (B) Fire: no flammable products — water and salt are produced. Option (D) GT: toxic gas arises when acids react with sulfides, cyanides, or other reactive anions — not from plain caustic NaOH. The compatibility chart is a required reference for hazardous waste treatment facility operators.

Handbook: Safety — Chemical Compatibility Chart (p.26): Group 1 (Acids, Minerals, Non-Oxidizing) + Group 10 (Caustics) → H (Heat generation) • Handbook p.26 — Chemical Compatibility Chart (Reactivity codes and group interactions)

Q11. [D12-Q11 • MCQ • Site characterization — coefficient of variation (CV)]

Soil samples collected from a contaminated site for a Phase II investigation yield an average arsenic concentration of $\bar{x} = 12.0$ mg/kg with a standard deviation of $s = 2.4$ mg/kg. The coefficient of variation (CV) for this data set is most nearly:

Coefficient of Variation (Handbook p.336): $CV = (100 \times s) / \bar{x}$

Relevant: $CV = \frac{100 \times s}{\bar{x}}$

- (A) 0.2%
- (B) **20.0%** ✓
- (C) 80.0%
- (D) 500%

Answer: (B)

$$CV = \frac{100 \times 2.4}{12.0} = \frac{240}{12.0} = 20.0\%$$

Option (A) 0.2%: divides without multiplying by 100 $\rightarrow CV = s/\bar{x} = 2.4/12.0 = 0.20$ (dimensionless fraction, not percent). Option (C) 80.0%: computes $(\bar{x} - s)/\bar{x} \times 100 = 9.6/12 \times 100 = 80\%$ — a confidence-interval-like value, not CV. Option (D) 500%: inverts the ratio $\rightarrow \bar{x}/s \times 100 = 12/2.4 \times 100 = 500\%$. CV is a normalized measure of dispersion used in DQO analysis to select sample sizes (see the Handbook sampling table, p.336). A CV of 20% is in the moderate range; CV $\geq 50\%$ would signal high variability requiring more samples.

Handbook: Environmental Engineering — Sampling and Monitoring: $CV = (100 \times s) / \bar{x}$ (p.336) • Handbook p.336 — Sampling and Monitoring: Coefficient of Variation

Q12. [D12-Q12 • MCQ • Site characterization — DQO sampling: number of samples required]

A planned remedial response site investigation has a coefficient of variation (CV) of 15%. The project requires a confidence level of 95% and a statistical power of 95%. The minimum detectable relative difference (MDRD) is set at 20%. Using the NCEES one-sided one-sample t-test table (Handbook p.336), the minimum number of samples required is most nearly:

- (A) 5
- (B) 6
- (C) 8 ✓
- (D) 12

Answer: (C)

Locate in the Handbook DQO table (p.336): row CV=15%, power=95%, confidence level=95%, column MDRD=20% → **n = 8**.

For reference, adjacent entries in the table: - MDRD=30% → n=5 (acceptable if a larger relative difference is tolerable) - MDRD=10% → n=26 (required if a smaller difference must be detected) - Confidence=99% (same CV, power, MDRD=20%) → n=12 — Option (D) - Confidence=90% (same CV, power, MDRD=20%) → n=6 — Option (B)

The DQO process sets the minimum sample size to achieve the desired statistical performance. Planned remedial response operations typically require 90–95% confidence and 90–95% power per EPA guidance; n=8 is the correct answer for the given parameters.

Handbook: Environmental Engineering — Sampling and Monitoring: DQO one-sided one-sample t-test table (p.336)

• Handbook p.336 — Sampling and Monitoring: Number of Samples Required table

Q13. [D12-Q13 • MCQ • MSW energy — composite heating value of mixed waste stream (USCS)]

A waste-to-energy facility receives a municipal solid waste stream with the following composition (by mass): 40% food wastes, 35% paper, and 25% plastics. Using Handbook typical heating values (food wastes 2,000 Btu/lb; paper 7,200 Btu/lb; plastics 14,000 Btu/lb), the composite as-fired heating value of this mixed stream is most nearly:

Relevant: $HV_{\text{composite}} = \sum_i f_i \cdot HV_i$

- (A) 5,020 Btu/lb
- (B) **6,820 Btu/lb ✓**
- (C) 7,733 Btu/lb
- (D) 8,620 Btu/lb

Answer: (B)

$$\begin{aligned} HV &= 0.40 \times 2,000 + 0.35 \times 7,200 + 0.25 \times 14,000 \\ &= 800 + 2,520 + 3,500 = 6,820 \text{ Btu/lb} \end{aligned}$$

Option (A) 5,020 Btu/lb: uses food waste HV for all components $\rightarrow$ 2000 Btu/lb scaled, or an arithmetic slip. Option (C) 7,733 Btu/lb: unweighted average $(2000+7200+14000)/3 = 7733$ — ignores the fact that food waste is the dominant fraction. Option (D) 8,620 Btu/lb: swaps fractions for food and plastics $\rightarrow 0.25 \times 2000 + 0.35 \times 7200 + 0.40 \times 14000 = 500 + 2520 + 5600 = 8620$. The high plastic fraction (25% at 14,000 Btu/lb) boosts the composite well above the all-MSW typical of 4,000–6,500 Btu/lb; the high food fraction (40% at only 2,000 Btu/lb) pulls it back down.

Handbook: Environmental Engineering — Municipal Solid Wastes (MSW): Typical Heating Value table; food 2,000 / paper 7,200 / plastics 14,000 / MSW composite 4,000–6,500 Btu/lb typical (p.359) • Handbook p.359 — MSW: Typical Heating Value of MSW Components table

Q14. [D12-Q14 • MCQ • RCRA — four hazardous waste characteristics (regulatory)]

Under the Resource Conservation and Recovery Act (RCRA), a solid waste is considered a listed or characteristic hazardous waste if it exhibits one or more of four regulatory characteristics. Which of the following correctly identifies all four RCRA hazardous waste characteristics?

- (A) **Ignitability, corrosivity, reactivity, and toxicity** ✓
- (B) Flammability, corrosivity, reactivity, and toxicity
- (C) Ignitability, corrosivity, reactivity, and radioactivity
- (D) Ignitability, corrosivity, persistence, and toxicity

Answer: (A)

The four RCRA hazardous waste characteristics (40 CFR Part 261, Subpart C) are: **ignitability** (D001 — flash point $\leq 60^{\circ}\text{C}/140^{\circ}\text{F}$ for liquids, or capable of causing fire), **corrosivity** (D002 — aqueous pH ≤ 2 or ≥ 12.5 , or corrodes steel at ≥ 6.35 mm/yr), **reactivity** (D003 — unstable, reacts violently with water, generates toxic gases, or is explosive), and **toxicity** (D004–D043 — determined by the Toxicity Characteristic Leaching Procedure, TCLP).

Option (B) Flammability: RCRA uses 'ignitability,' not 'flammability' — the regulatory term matters for permits and manifests. Option (C) Radioactivity: radioactive wastes are regulated separately under the Atomic Energy Act (AEA/NRC) and DOE orders — not as a RCRA characteristic. Option (D) Persistence: environmental persistence is relevant to listing criteria and PBT chemicals but is not one of the four characteristic tests. The TCLP test for toxicity leaches a representative sample under simulated landfill conditions and compares extract concentrations to regulatory thresholds.

Handbook: Not in Handbook v10.6 — RCRA hazardous waste characteristics are regulatory knowledge (40 CFR 261) • RCRA 40 CFR Part 261 Subpart C — four hazardous waste characteristics (ignitability, corrosivity, reactivity, toxicity)

Q15. [D12-Q15 • MCQ • Solid waste — diversion rate (SI)]

A municipality generates 5.0 tonnes/day of municipal solid waste (MSW). Of this, 1.5 tonnes/day is diverted from landfill through recycling and composting programs. The solid waste diversion rate for this municipality is most nearly:

Relevant: Diversion Rate = $\frac{\text{mass diverted from landfill}}{\text{total mass generated}} \times 100\%$

- (A) 23.1%
- (B) **30.0%** ✓
- (C) 42.9%
- (D) 70.0%

Answer: (B)

$$\text{Diversion Rate} = \frac{1.5}{5.0} \times 100\% = 30.0\%$$

Option (A) 23.1%: uses total as denominator including diverted + landfilled + generated together $\rightarrow 1.5/(5.0+1.5) = 1.5/6.5 = 23.1\%$ — double-counts material. Option (C) 42.9%: uses landfilled mass as denominator $\rightarrow 1.5/(5.0-1.5) = 1.5/3.5 = 42.9\%$ — rates diverted relative to landfilled, not total. Option (D) 70.0%: computes the landfill disposal rate ($1 - 0.30 = 0.70 = 70\%$) — this is the complement of the diversion rate. The EPA has set a national goal to divert 50% of MSW from landfills through source reduction and recycling. The U.S. recycling/composting rate was approximately 32% as of recent EPA data.

Handbook: Fundamental definition: Diversion Rate = diverted mass / total generated mass $\times 100\%$ (not a named Handbook formula) • Fundamental: solid waste diversion rate definition

Q16. [D12-Q16 • MCQ • RCRA — minimum DRE standard for hazardous waste incineration]

Under RCRA (40 CFR 63.1206), a permitted hazardous waste incinerator must meet a minimum Destruction and Removal Efficiency (DRE) for each principal organic hazardous constituent (POHC) in the waste feed. The RCRA minimum DRE standard for POHCs is:

$DRE = (W_{in} - W_{out}) / W_{in} \times 100\%$ [Handbook p.325]

Relevant: $DRE = \frac{W_{in} - W_{out}}{W_{in}} \times 100\%$

- (A) 99.0%
- (B) 99.9%
- (C) **99.99%** ✓
- (D) 99.9999%

Answer: (C)

The RCRA standard requires $DRE \geq 99.99\%$ (four nines) for each POHC during a trial burn. This means the incinerator must destroy at least 9,999 out of every 10,000 kg of POHC fed to it — $W_{out} \leq 0.01\%$ of W_{in} . For especially toxic constituents — dioxins and furans — EPA imposes a more stringent standard of 99.9999% (six nines).

Option (A) 99.0%: this is the general combustion efficiency (CE) range for well-designed combustors but not the RCRA DRE threshold. Option (B) 99.9%: three nines — not the RCRA standard; this shortfall allows 10× more POHC to escape than the regulation permits. Option (D) 99.9999%: the dioxin/furan DRE standard — more stringent than the general POHC requirement. The trial burn protocol includes stack sampling for the designated POHCs; failure to meet DRE during trial burn means the facility cannot receive an operating permit.

Handbook: Environmental Engineering — Incineration: $DRE \text{ formula} = (W_{in} - W_{out}) / W_{in} \times 100\%$ (p.325); RCRA 99.99% standard is regulatory knowledge (40 CFR 63.1206) • Handbook p.325 — DRE formula; RCRA 40 CFR 63 — 99.99% DRE standard for POHCs

Q17. [D12-Q17 • Numeric • Landfill — clay liner leachate breakthrough time, second scenario (USCS)]

A proposed industrial solid waste landfill uses a 3-ft compacted clay liner with hydraulic conductivity $K = 0.075$ ft/yr and porosity $\eta = 0.25$. The design hydraulic head above the liner is $h = 2$ ft. Calculate the leachate breakthrough time for the clay liner (in years).

Break-Through Time (Handbook p.333): $t = d^2\eta / [K(d + h)]$

Relevant: $t = \frac{d^2\eta}{K(d + h)}$

Numeric entry.

Answer: 6.0 yr

$$t = \frac{(3 \text{ ft})^2 \times 0.25}{0.075 \text{ ft/yr} \times (3 + 2) \text{ ft}} = \frac{9 \times 0.25}{0.075 \times 5} = \frac{2.25}{0.375} = 6.0 \text{ yr}$$

Compare with Q01 ($d=2$ ft, $K=0.10$ ft/yr $\rightarrow t=2.0$ yr): the 3-ft liner in this problem is 50% thicker, which increases t by a factor of $(3/2)^2=2.25$ in the numerator, but the lower $K=0.075$ ft/yr (vs 0.10) further reduces the denominator. The combined effect: $(9 \times 0.25)/(0.075 \times 5) = 2.25/0.375 = 6.0$ yr — $3 \times$ longer than the 2-ft liner. Modern Subtitle D landfills require $K \leq 1 \times 10^{-7}$ cm/s (≈ 0.000105 ft/yr) for clay liners, giving breakthrough times of thousands of years for even modest liner thicknesses.

Handbook: Environmental Engineering — Landfill: Break-Through Time $t = d^2\eta/[K(d+h)]$ (p.333) • Handbook p.333
— Landfill: Leachate Breakthrough Time

Q18. [D12-Q18 • MCQ • Hazardous waste — TCLP toxicity characteristic determination (in-stem)]

A solid waste generator performs TCLP (Toxicity Characteristic Leaching Procedure) extraction on a sample of industrial sludge. The regulatory threshold for lead under TCLP is 5.0 mg/L (as given by 40 CFR 261.24 and stated here for reference). The TCLP extract analysis reports a lead concentration of 8.0 mg/L. Which conclusion is correct regarding this waste?

- (A) The waste is NOT hazardous because the lead concentration exceeds the TCLP threshold.
- (B) **The waste IS hazardous because the lead concentration exceeds the TCLP threshold. ✓**
- (C) The waste IS hazardous only if it also exhibits ignitability, corrosivity, or reactivity.
- (D) The waste IS NOT hazardous because 8.0 mg/L is below the TCLP threshold.

Answer: (B)

The measured TCLP extract concentration (8.0 mg/L) ****exceeds**** the regulatory threshold (5.0 mg/L). Therefore, the waste exhibits the toxicity characteristic (D008 for lead) and ****is a RCRA characteristic hazardous waste****.

Option (A): Incorrect statement — 'not hazardous because it exceeds' is contradictory; exceeding the threshold makes it hazardous, not safe. Option (C): A waste is a RCRA characteristic hazardous waste if it fails ANY of the four characteristic tests independently — it does not need to exhibit multiple characteristics simultaneously. Option (D): Incorrect — 8.0 mg/L > 5.0 mg/L (threshold is not exceeded in the safe direction). Once classified as hazardous, this sludge must be managed under RCRA Subtitle C requirements: proper containers, manifesting, permitted storage, and disposal at a licensed TSDF (Treatment, Storage, and Disposal Facility).

Handbook: Not in Handbook v10.6 — TCLP threshold for Pb (5.0 mg/L) is given in the problem stem per 40 CFR 261.24 • RCRA 40 CFR 261.24 — Toxicity Characteristic (TCLP); threshold stated in problem stem

Q19. [D12-Q19 • MCQ • Hazardous waste — chemical compatibility (organic acid + amines)]

A hazardous waste storage area contains drums of organic acids (Group 3 — e.g., acetic acid) and drums of aliphatic amines (Group 7 — e.g., triethylamine). According to the NCEES Chemical Compatibility Chart (Handbook p.26), what hazards result if these two groups are accidentally mixed?

Reactivity codes: H = Heat generation · GT = Toxic gas generation · GF = Flammable gas · F = Fire · E = Explosion · G = Innocuous & non-flammable gas

- (A) G — Innocuous and non-flammable gas generation only
- (B) E — Explosion
- (C) H + F — Heat generation and fire
- (D) **H + GT — Heat generation and toxic gas generation ✓**

Answer: (D)

From the Chemical Compatibility Chart (p.26): Group 3 (Acids, Organic) × Group 7 (Amines, Aliphatic & Aromatic) → **H (Heat generation) + GT (Toxic gas generation)**.

Organic acid + amine undergoes acid–base neutralization: $\text{R-COOH} + \text{R}_3\text{N} \rightarrow \text{R-COO}^- \text{R}_3\text{NH}^+$ (exothermic → H). If the amine is volatile, vapors can be released and, depending on the specific amine, may be toxic (GT — e.g., triethylamine TLV = 1 ppm TWA, ammonia from simple amines). Option (A) G: innocuous gas — acids and amines do NOT produce CO_2 or N_2 harmlessly; the products include heat and potentially toxic amine vapors. Option (B) E: explosion code not applicable to simple acid–base neutralization. Option (C) H + F: fire code (F) is not assigned; amines are flammable but the reaction itself is not primarily a fire hazard — the flash point of the amine vapor, not the reaction, drives fire risk. Chemical compatibility assessment is required by OSHA 1910.120 for hazardous waste operations.

Handbook: Safety — Chemical Compatibility Chart (p.26): Group 3 (Acids, Organic) + Group 7 (Amines, Aliphatic & Aromatic) → H + GT • Handbook p.26 — Chemical Compatibility Chart

Q20. [D12-Q20 • MCQ • Landfill — daily airspace consumption from mass balance (USCS)]

A municipal solid waste landfill receives 500 tons/day of compacted waste. The as-placed density of the compacted waste in the landfill (including daily cover) is 1,500 lb/yd³. The volume of landfill airspace consumed per day is most nearly:

Fundamental mass balance: $V = \text{mass} / \text{density}$

$$\text{Relevant: } V = \frac{\dot{m}}{\rho} = \frac{(\text{tons/day} \times 2,000 \text{ lb/ton})}{\rho \text{ (lb/yd}^3\text{)}}$$

- (A) 0.33 yd³/day
- (B) 375 yd³/day
- (C) **667 yd³/day** ✓
- (D) 1,000 yd³/day

Answer: (C)

$$V = \frac{500 \text{ tons/day} \times 2,000 \text{ lb/ton}}{1,500 \text{ lb/yd}^3} = \frac{1,000,000 \text{ lb/day}}{1,500 \text{ lb/yd}^3} = 667 \text{ yd}^3/\text{day}$$

Option (A) 0.33 yd³/day: forgets the 2000 lb/ton conversion $\rightarrow 500/1500 = 0.33$ — orders of magnitude off. Option (B) 375 yd³/day: inverts density in conversion $\rightarrow 500 \times 1500 / 2000 = 375$ — uses density as numerator instead of denominator. Option (D) 1,000 yd³/day: uses a placed density of 1,000 lb/yd³ instead of 1,500 — a common order-of-magnitude but off by 50%. At 667 yd³/day, this landfill would exhaust 1 million yd³ of airspace in about 4.1 years. Landfill airspace calculations drive the permitted life of a facility and are central to solid waste planning and permitting.

Handbook: Fundamental mass balance: $V = \dot{m} / \rho$ — conservation of mass (not a named Handbook formula; Handbook provides densities and compaction formulas on p.332–334) • Fundamental: landfill airspace = (mass rate $\times$ unit conversion) / placed density